

Assignment 1: Data Cleaning and Summarising

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Data Preparation

The dataset provided for the assignment is a Global database on school-age digital connectivity. The dataset consists of 11 columns and 90 rows. Before doing any analysis on the dataset, the data is prepared and cleaned to minimize/remove the errors in the dataset so that it becomes consistent.

Error Type 1: Incorrect data type (Percentage Format Conversion)

The columns Total, Residence (Rural), Residence (Urban), Wealth quintile (Poorest) and Wealth quintile (Richest) in the dataset contains percentage values. These columns were stored as string data type. The "%" sign was stripped and then the remaining the string was converted into a float data type. Afterwards the values were divided by 100 to have a suitable format for analysis. This type conversion was essential since it allows us to perform mathematical operations and statistical analysis. Without conversion, we might have resulted in errors or misleading values.

Error Type 2: Handling Missing Values

In the dataset, columns Residence (Rural), Residence (Urban), Wealth quintile (Poorest), and Wealth quintile (Richest) had 5, 4, 7 and 7 missing values respectively. The rows having missing values for these columns were dropped. This approach was taken because the number of missing values was small and removing them would not significantly reduce the size of the dataset. By dropping missing values, it was ensured that the analysis won't be biased because of incomplete data entry.

Error Type 3: Outlier Detection and Removal

To visually check the data for outliers, box plots were utilized. For this, box plots are helpful because they display the data distribution clearly and draw attention to any values that deviate from the norm (outliers). Values above the predicted range for percentages (i.e., values greater than 1.0) were found in the dataset. The time-period column in the dataset had some implausible values like year 2076, 2099 and 3562, these year values were fixed. Since outliers can skew statistical measures like the mean and standard deviation etc. and provide false results, hence these rows were dropped.

Error Type 4: Removing Duplicate Rows

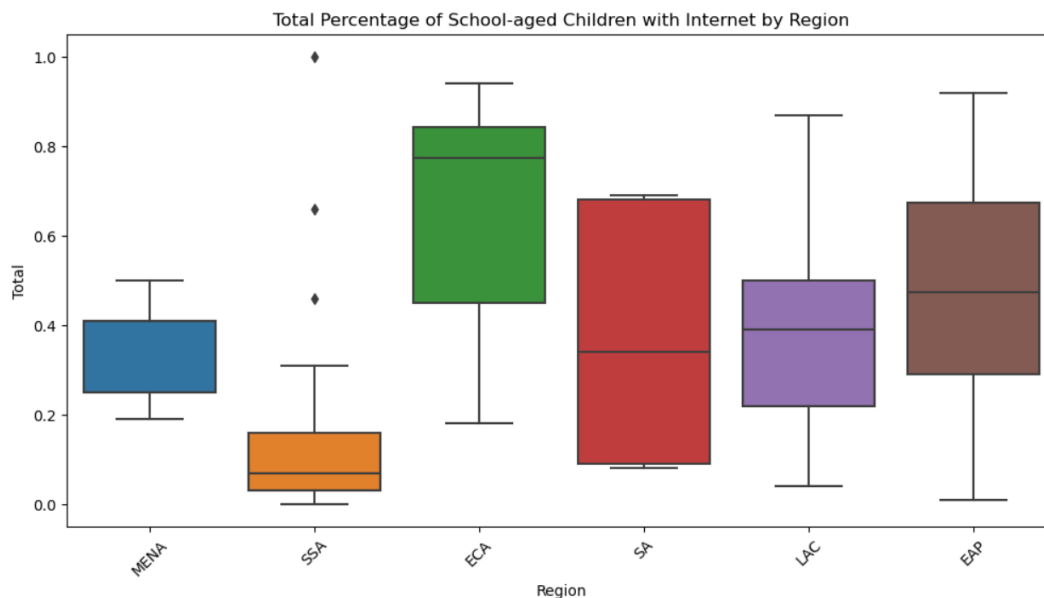
If duplicate rows in datasets are ignored, they may result in inaccurate analysis and findings. Data entry errors and dataset merging are two common causes of duplicate rows. After examining the dataset for duplicates in every column, three rows were found and eliminated. Eliminating duplicates enhances the precision and dependability of your research and is a typical data-cleaning procedure. Making sure that every row in the dataset is distinct and prevent the values/results from being skewed.

Data Exploration

Task 2.1

The objective of this task was understanding the spread of internet connectivity among school-age children across various locations.

The distribution of the total percentage of school-age children with internet connection at home, categorized by region, was visualized using box plots. Following that, the median percentage which offers a central measure of the distribution was computed for each region.



Median percentage by region:

Region	Median %
EAP	0.475
ECA	0.775
LAC	0.390
MENA	0.410
SA	0.340
SSA	0.070

Findings: With a median internet access score of 0.775, **Eastern Europe and Central Asia** (ECA) has the highest level of connection. The region of **East Asia and the Pacific** (EAP) and the **Middle East and North Africa** (MENA), exhibit moderate access, with median values of roughly 0.475 and 0.410, respectively. The regions of **Latin America and the Caribbean** (LAC) and for **South Asia** (SA) show lower access, with median values of **0.390** and **0.340**. The region of **Sub-Saharan Africa** (SSA) has the lowest median (0.07), indicating significant barriers to internet access.

Task 2.2

Comparing internet connectivity between the wealthiest and poorest quintiles and emphasizing the top nations in each category was the aim of this task.

The mean percentage of school-aged children with internet access at home for both the Poorest and Richest wealth quintiles was calculated. Afterwards the top 10 countries with the highest percentages in each category were identified.

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Mean percentage for Poorest wealth quintile: 0.18
Mean percentage for Richest wealth quintile: 0.61
Top 10 countries for Poorest wealth quintile:
  Countries and areas  Wealth quintile (Poorest)
59          Somalia          1.00
8           Brazil          0.84
68          Tonga          0.83
14          Chile          0.75
61         Sri Lanka          0.71
66   North Macedonia          0.68
57          Serbia          0.65
33          Japan          0.64
37         Kyrgyzstan          0.56
45         Montenegro          0.55
Top 10 countries for Richest wealth quintile:
  Countries and areas  Wealth quintile (Richest)
4           Barbados          1.00
9           Bulgaria          1.00
57          Serbia          1.00
59          Somalia          1.00
66   North Macedonia          1.00
2           Armenia          0.99
14          Chile          0.99
16          Colombia          0.99
17         Costa Rica          0.99
25          Georgia          0.99

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Findings:

Poorest Wealth Quintile: The most economically disadvantaged individuals have minimal connectivity, as indicated by the mean percentage of internet access for this quintile, which is 0.18 (18%). Among the poorest nations, Somalia has complete internet access (1.00 or 100%), which sets it apart from other nations. Brazil (0.84), Tonga (0.83), and Chile (0.75) trail, suggesting improved accessibility even for the most impoverished groups. The existence of nations like Japan (0.64) and Sri Lanka (0.71) indicates differing degrees of success in giving the poorest people access to the internet.

Richest Wealth Quintile: The quintile with the highest mean percentage, 0.61 (61%), indicates a notably higher level of access in the wealthiest groups. Barbados, Bulgaria, Serbia, Somalia, and North Macedonia are in the richest quintile of nations with full internet access (1.00 or 100%). Other nations that are nearly at full access (0.99 or 99%) include Armenia, Chile, Colombia, Costa Rica, and Georgia.

Task 2.3

Comparing internet connectivity in rural and urban locations within the Lower Middle-Income group was the aim of this task.

The dataset's mean, median, and standard deviation were then determined for the Residence (Rural) and Residence (Urban) columns after it was filtered only to contain the Lower Middle-Income group. These statistics show how internet connectivity is distributed and varies in certain locations.

Findings:

Rural - Mean: 0.12, Median: 0.04, Std Dev: 0.16

Urban - Mean: 0.28, Median: 0.23, Std Dev: 0.21

Rural Area: With a mean percentage of 0.12(12%) in rural areas, there is severely limited connectivity in these locations. The median number is significantly lower, at 0.04 (4%), indicating that less than 4% of school-age children in more than half of the rural areas have access to the internet at home. Although there is some variation in rural internet connectivity, overall availability in these locations is still poor, as seen by the standard deviation of 0.16.

Urban Area: By comparison, the urban mean is 0.28 (28%), which is more than twice as high as the rural mean. This demonstrates how much better internet connectivity is in metropolitan areas. With almost 25% of the population online, the median of 0.23 (23%) indicates that at least half of urban regions have comparatively better connectivity. With some urban locations having far greater connectivity than others, the 0.21 standard deviation in urban regions suggests increased unpredictability and a wider range of internet access levels.

There is a clear difference between rural and urban areas; on average, metropolitan areas have access to over twice as much internet as rural areas.

Use of AI Tools

Generative AI tools like ChatGPT and Val were helpful while doing task 1 and 2 as whenever an error was encountered, which I couldn't fix on my own, I asked for help with these errors. The bot helped me point out the errors and then I could fix those. Apart from there was nothing else to use AI tools for since the assignment was straightforward, and course resources were enough to complete the assignment.

References

Course recourses used:

https://rmit.instructure.com/courses/125162/files/40284486/download?download_frd=1

https://rmit.instructure.com/courses/125162/files/39724363/download?download_frd=1

https://rmit.instructure.com/courses/125162/files/40379180/download?download_frd=1

https://rmit.instructure.com/courses/125162/files/39724010/download?download_frd=1

Other Resources:

https://matplotlib.org/stable/plot_types/stats/boxplot_plot.html#sphx-glr-plot-types-stats-boxplot-plot-py